Need of Data Analytics:

Now that you have looked at what data analytics is, let's understand how we can use data analytics.

Fig: Ways to use Data Analytics

1**. Improved Decision Making:** Data Analytics eliminates guesswork and manual tasks. Be it choosing the right content, planning marketing campaigns, or developing products. Organizations can use the insights they gain from data analytics to make informed decisions. Thus, leading to better outcomes and customer satisfaction.

2. **Better Customer Service:** Data analytics allows you to tailor customer service according to their needs. It also provides personalization and builds stronger relationships with customers. Analyzed data can reveal information about customers' interests, concerns, and more. It helps you give better recommendations for products and services.

3. **Efficient Operations**: With the help of data analytics, you can streamline your processes, save money, and boost production. With an improved understanding of what your audience wants, you spend lesser time creating ads and content that aren't in line with your audience's interests.

4**. Effective Marketing**: Data analytics gives you valuable insights into how your campaigns are performing. This helps in fine-tuning them for optimal outcomes. Additionally, you can also find potential customers who are most likely to interact with a campaign and convert into leads.

Let’s now dive into the various steps involved in data analytics.

Data Analytics Process and Tools:

Steps Involved in Data Analytics:

There are a few steps that are involved in the data analytics lifecycle. Let’s have a look at it with the help of an analogy.

Imagine you are running an e-commerce business and your company has nearly a million in customer base. Your aim is to figure out certain problems related to your business, and subsequently come up with data-driven solutions to grow your business.

Below are the steps that you can take to solve your problems.

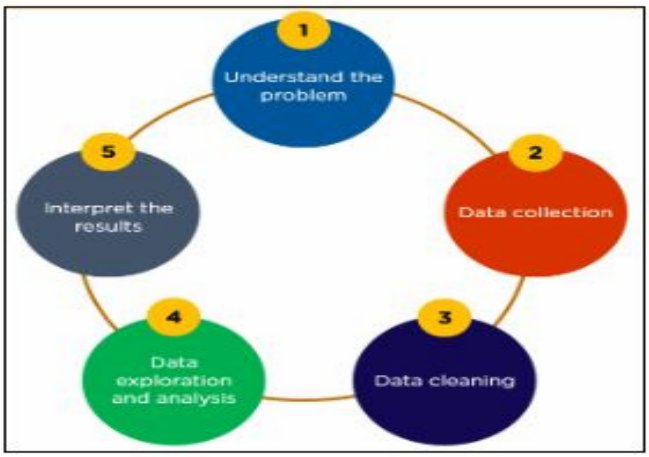

Fig: Data Analytics process steps

1**. Understand the problem:** Understanding the business problems, defining the organizational goals, and planning a lucrative solution is the first step in the analytics process. E-commerce companies often encounter issues such as predicting the return of items, giving relevant product recommendations, cancellation of orders, identifying frauds, optimizing vehicle routing, etc.

2. **Data Collection:** Next, you need to collect transactional business data and customer-related information from the past few years to address the problems your business is facing. The data can have information about the total units that were sold for a product, the sales, and profit that were made, and also when was the order placed. Past data plays a crucial role in shaping the future of a business.

3. **Data Cleaning:** Now, all the data you collect will often be disorderly, messy, and contain unwanted missing values. Such data is not suitable or relevant for performing data analysis. Hence, you need to clean the data to remove unwanted, redundant, and missing values to make it ready for analysis.

4. **Data Exploration and Analysis**: After you gather the right data, the next vital step is to execute exploratory data analysis. You can use data visualization and business intelligence tools, data mining techniques, and predictive modeling to analyze, visualize, and predict future outcomes from this data. Applying these methods can tell you the impact and relationship of a certain feature as compared to other variables.

Below are the results you can get from the analysis:

- You can identify when a customer purchases the next product.
- You can understand how long it took to deliver the product.
- You get a better insight into the kind of items a customer looks for, product returns, etc.
- You will be able to predict the sales and profit for the next quarter.
- You can minimize order cancellation by dispatching only relevant products.
- You'll be able to figure out the shortest route to deliver the product, etc.

5. **Interpret the results:** The final step is to interpret the results and validate if the outcomes meet your expectations. You can find out hidden patterns and future trends. This will help you gain insights that will support you with appropriate data-driven decision making.

Data Analytics Tools

Now that we looked at the different steps involved in data analytics, let's see the tools involved in data analytics, to perform the above steps. In this blog, we will discuss 7 data analytics tools, including a couple of programming languages that can help you perform analytics better.

Fig: Data Analytics for Beginners - Tools used

1. Python: Python is an object-oriented open-source programming language. It supports a range of libraries for data manipulation, data visualization, and data modeling.

2. R: R is an open-source programming language majorly used for numerical and statistical analysis. It provides a range of libraries for data analysis and visualization.

3. Tableau: It is a simplified data visualization and analytics tool. This helps you create a variety of visualizations to present the data interactively, build reports, and dashboards to showcase insights and trends.

4. Power BI: Power BI is a business intelligence tool that has an easy 'drag and drop functionality. It supports multiple data sources with features that visually appeal to

data. Power BI supports features that help you ask questions to your data and get immediate insights.

5. QlikView: QlikView offers interactive analytics with in-memory storage technology to analyze vast volumes of data and use data discoveries to support decision making. It provides social data discovery and interactive guided analytics. It can manipulate colossal data sets instantly with accuracy.

6. Apache Spark: Apache Spark is an open-source data analytics engine that processes data in real-time and carries out sophisticated analytics using SQL queries and machine learning algorithms.

7. SAS: SAS is a statistical analysis software that can help you perform analytics, visualize data, write SQL queries, perform statistical analysis, and build machine learning models to make future predictions.

Now that you have seen the data analytics tools, let’s jump ahead and see the applications of data analytics.

Data Analytics Applications

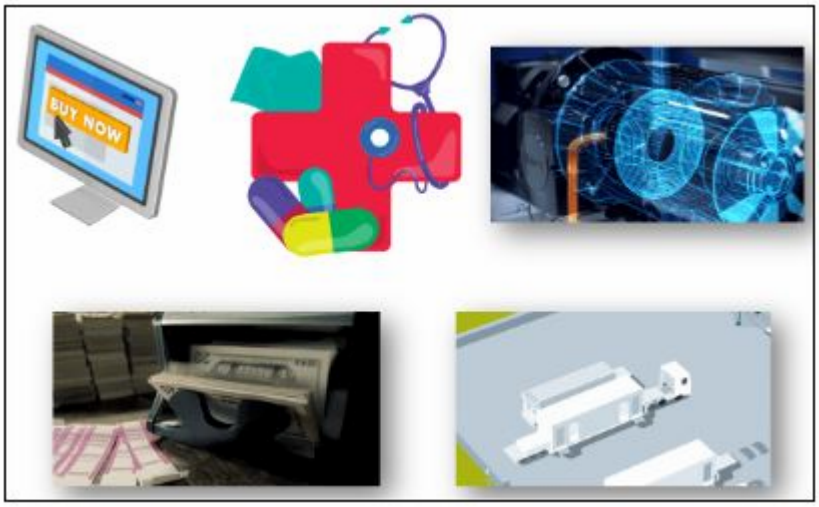

Fig: Various applications of data analytics

Data analytics is used in almost every sector of business, let's discuss a few of them:

1. Retail: Data analytics helps retailers understand their customer needs and buying habits to predict trends, recommend new products, and boost their business.

They optimize the supply chain, and retail operations at every step of the customer journey.

2. Healthcare: Healthcare industries analyze patient data to provide lifesaving diagnoses and treatment options. Data analytics help in discovering new drug development methods as well.

3. Manufacturing: Using data analytics, manufacturing sectors can discover new cost-saving opportunities. They can solve complex supply chain issues, labor constraints, and equipment breakdowns.

4. Banking sector: Banking and financial institutions use analytics to find out probable loan defaulters and customer churn out rate. It also helps in detecting fraudulent transactions immediately.

5. Logistics: Logistics companies use data analytics to develop new business models and optimize routes. This, in turn, ensures that the delivery reaches on time in a cost-efficient manner.